

-- Cox Regression - Sarcopenia Onset -----

i Definition : ewgsop2
i Covariates : fixed
i Dairy : continuous
i Route : cc
i Interaction: none

-- Building survival dataset (CC) --

Events: 38 sarcopenia cases out of 917 participants

-- Fixed covariates dataset (one row per pt) --

??? Fixed dataset: 917 rows | 35 events

-- Fitting Cox models (CC) --

Unadjusted formula: survival::Surv(time, event) ~ dairy_exposure
Adjusted formula : survival::Surv(time, event) ~ dairy_exposure + BMI +
education_level + and smoking_status + mvpa_min_day_f1 + diabetes_status

-- Events Per Variable (EPV) [cc]

i Events : 35
i Coefs : 8
i EPV : 4.4

-- 1. Proportional Hazards - cox.zph() | pass: global p > 0.05

```
chisq df p dairy_exposure 0.07585 1 0.783 BMI 0.00364 1 0.952 education_level
2.87448 2 0.238 smoking_status 7.79096 2 0.020 mvpa_min_day_f1 0.54652 1 0.460
diabetes_status 1.71154 1 0.191 GLOBAL 14.31768 8 0.074
??? PH assumption not violated (global p = 0.0739)
Schoenfeld plots saved to
03_outputs/Cox/cc_ewgsop2_fixed_continuous_27072026_1237
```

-- 2. Linearity - martingale (null model) vs continuous predictors | pass: smooth ??? linear

```
'geom_smooth()' using formula = 'y ~ x'
'geom_smooth()' using formula = 'y ~ x'
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Linearity plots saved to
03_outputs/Cox/cc_ewgsop2_fixed_continuous_27072026_1237
```

-- 3. Outliers - deviance & martingale residuals | pass: |dev| <= 3

i 2 observation(s) with |deviance residual| > 3.

-- 4. Influential observations - std. DFBetas | pass: no extreme spikes

i 24 obs exceed |DFBeta_std| > 0.2.

-- 5. Overall Fit - Cox-Snell residuals | pass: points on 45° diagonal

```
Cox-Snell plot saved to
03_outputs/Cox/cc_ewgsop2_fixed_continuous_27072026_1237
```

-- 6. Multicollinearity - VIF | pass: VIF < 5 (< 10 relaxed)

```
GVIF Df GVIF^(1/(2*Df)) dairy_exposure 1.086 1 1.042 BMI 1.260 1 1.122
education_level 1.383 2 1.084 smoking_status 1.182 2 1.043 mvpa_min_day_f1
1.094 1 1.046 diabetes_status 1.343 1 1.159
```

-- Incidence rates by dairy quartile [cc]

quartile	n_pts	n_events	person_years	rate_per_1000py	ci_low	ci_high	1	Q1 (low)
230	9	162	03.4	0.56	0.25	1.05	2	Q2
229	6	160	71.2	0.37	0.14	0.81	3	Q3
229	9	161	22.3	0.56	0.26	1.06	4	Q4 (high)
229	11	163	11.8	0.67	0.34	1.21		

Incidence table saved to

03_outputs/Cox/cc_ewgsop2_fixed_continuous_27072026_1237

Model results saved to

03_outputs/Cox/cc_ewgsop2_fixed_continuous_27072026_1237/cc_model_results.csv